

Multiple Choice

1. **Answer:** A

Options: A) eight weeks post-fertilization.; B) fourteen weeks post-fertilization.; C) seven weeks post-fertilization.; D) twelve weeks post-fertilization.; E) four weeks post-fertilization.

Note: The palatine processes begin to fuse at eight weeks post-fertilization, which is a critical developmental milestone in the formation of the secondary palate, essential for proper oral and facial structure.

2. **Answer:** A

Options: A) They have an arranged marriage; B) They agree on almost everything; C) They have a strong religious belief; D) They maintain separate social lives; E) They always take vacations together

Note: The correct answer is supported by research indicating that arranged marriages often involve family support and shared values, which can contribute to greater marital stability and longevity compared to marriages based solely on romantic love.

3. **Answer:** B

Options: A) An opening in the anterior chest wall.; B) An opening in the anterior wall of the trachea below the cricoid cartilage.; C) An opening in the anterior chest wall, above the diaphragm.; D) An opening in the posterior wall of the esophagus.; E) An opening in the posterior wall of the trachea above the cricoid cartilage.

Note: A tracheostomy is defined as a surgical procedure that creates an opening in the anterior wall of the trachea, typically performed below the cricoid cartilage to establish an airway in cases of obstruction or respiratory distress.

4. **Answer:** A

Options: A) arise from the ascending aorta and do not fill during either systole or diastole.; B) arise from the arch of the aorta and do not fill during either systole or diastole.; C) arise from the ascending aorta and fill during systole.; D) arise from the arch of the aorta and fill during systole.; E) arise from the pulmonary artery and fill during systole.

Note: The coronary arteries arise from the ascending aorta, and during the cardiac cycle, they primarily fill with blood during diastole when the heart muscle relaxes, but the phrasing of the statement indicates a misunderstanding, as they do fill during diastole, making the correct answer misleading; the correct understanding is that they are perfused mainly during diastole due to the backflow from the aorta as the heart relaxes.

5. **Answer:** E

Options: A) Lp(a); B) HDL; C) ApoB-100; D) IDL; E) VLDL

Note: VLDL, or very-low-density lipoprotein, is the primary lipoprotein secreted by

the liver that transports triglycerides, which are often derived from dietary fats, to peripheral tissues for energy utilization and storage.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The patellar tap test is true because it effectively evaluates the presence of excess fluid in the knee joint by detecting the sensation of the patella moving against the underlying structures when tapped, indicating an effusion.

7. **Answer:** True

Options: A) True; B) False

Note: Case-control studies are effective in establishing associations between exposures, such as sugar intake, and outcomes, like dental caries, by comparing the histories of individuals with and without the condition, thus providing valuable insights into potential risk factors.

8. **Answer:** False

Options: A) True; B) False

Note: An enlarged submandibular salivary gland can often be palpated during a physical examination, as it may present as a noticeable swelling or mass in the submandibular area.

9. **Answer:** False

Options: A) True; B) False

Note: Using a stethoscope is not essential for evaluating the jugular venous pulse because this assessment can be accurately performed through visual observation and palpation of the neck without the need for acoustic amplification.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the notation "Asp 235 Phe" indicates that at position 235 of the protein sequence, the amino acid aspartic acid (Asp) has been substituted by phenylalanine (Phe), which is a common way to denote amino acid mutations in molecular biology.

Long Form Response

11. **Answer:** The emergence of the Middle East Respiratory Syndrome (MERS) coronavirus has significantly impacted global health due to its transmission dynamics, high mortality rate, and the public health responses it necessitated. MERS is primarily transmitted through direct contact with infected camels or through close human-to-human interaction, which complicates containment efforts. The mortality rate for

MERS has been alarmingly high, estimated at around 34%, leading to heightened concerns among health authorities worldwide. In response, global health organizations, including the World Health Organization, have implemented surveillance measures, enhanced diagnostic capabilities, and promoted public awareness campaigns to mitigate the spread. These efforts underscore the importance of strengthening public health infrastructure to respond effectively to emerging infectious diseases.

Note: correct because it accurately identifies MERS's unique transmission methods, its significantly high mortality rate, and the resulting public health responses required to manage its spread, all of which are critical factors in assessing its impact on global health.

12. **Answer:** Glucose-6-phosphate dehydrogenase (G₆PD) deficiency is primarily inherited in an X-linked recessive pattern, which means that the gene responsible for this enzyme deficiency is located on the X chromosome. As a result, males, who have only one X chromosome, are more severely affected by the deficiency, often presenting with hemolytic anemia when exposed to certain oxidative stressors. Females, possessing two X chromosomes, may be carriers and display milder symptoms or none at all, owing to X-inactivation. This genetic pattern has significant implications for affected populations, particularly in areas where malaria is prevalent, as G₆PD deficiency confers some protection against severe malaria. Understanding the inheritance and population dynamics of G₆PD deficiency is crucial for public health strategies and genetic counseling.

Note: The answer correctly identifies the X-linked recessive inheritance pattern of G₆PD deficiency, explaining its greater impact on males due to their single X chromosome and the potential for females to be carriers with milder or no symptoms, which highlights the genetic basis and implications for affected individuals and populations.